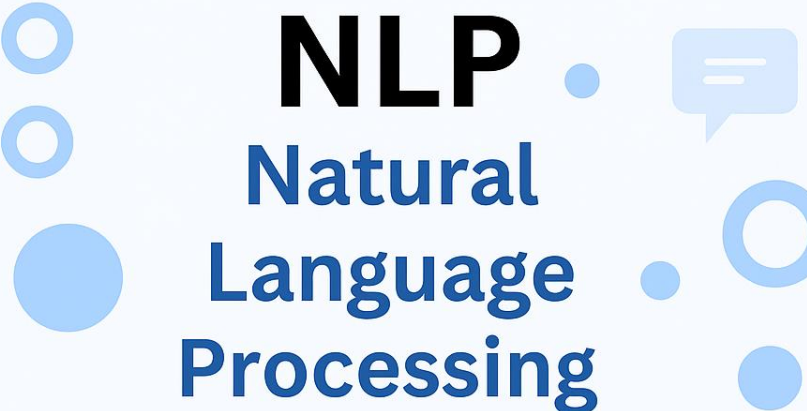
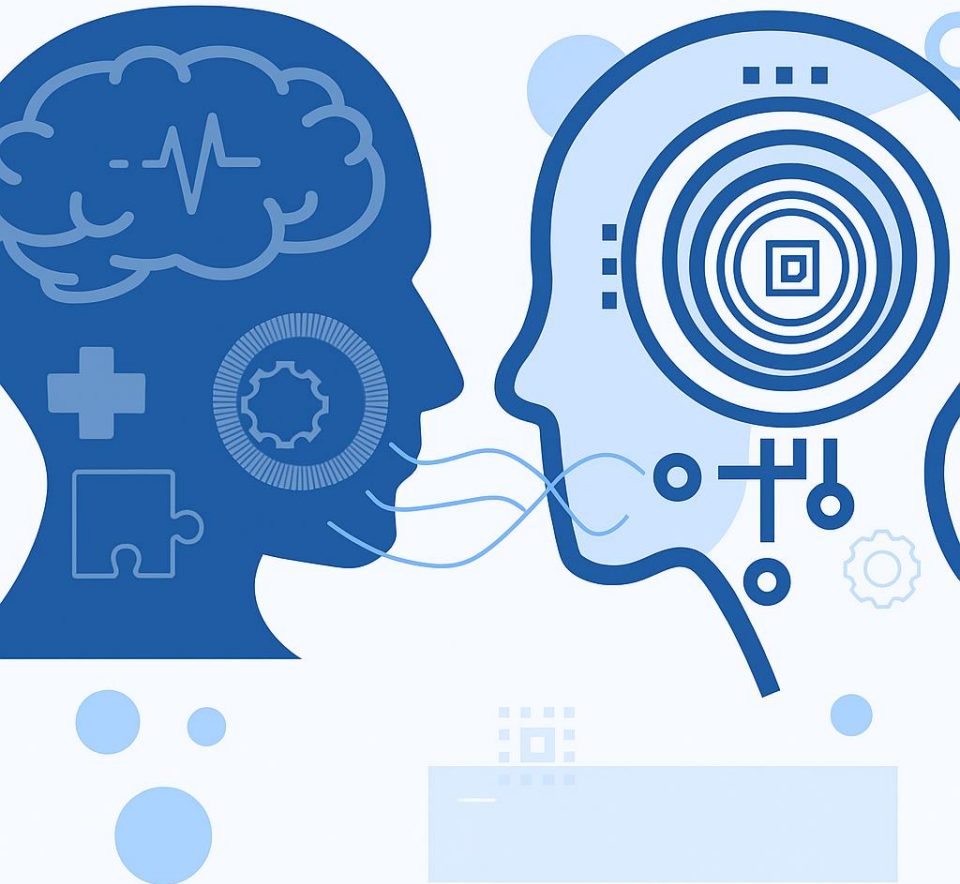




NLP

Natural
Language
Processing



NATURAL LANGUAGE PROCESSING (NLP)

PMDS606L

MODULE 4

LECTURE 1

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PART OF SPEECH (POS)

POS

Word Classes

Morphological Classes

Lexical Tags

PART OF SPEECH (POS)

POS	Example Words	Role in Sentence
Noun (NN)	dog, teacher, book	Names a person, place, thing, or idea
Verb (VB)	run, eat, think	Expresses an action or state
Adjective (JJ)	big, beautiful, quick	Describes a noun
Adverb (RB)	quickly, very, well	Modifies a verb, adjective, or another adverb
Pronoun (PRP)	he, she, it, they	Replaces a noun
Preposition (IN)	in, on, at, with	Shows relationship between words
Conjunction (CC)	and, but, or	Joins words or clauses
Interjection (UH)	oh!, wow!, hey!	Expresses emotion or surprise

Universal POS Tagset (NLP Standard)

1. ADJ – Adjective
2. ADP – Adposition (preposition/postposition)
3. ADV – Adverb
4. AUX – Auxiliary verb
5. CCONJ – Coordinating conjunction
6. DET – Determiner
7. INTJ – Interjection
8. NOUN – Common noun
9. NUM – Numeral
10. PART – Particle
11. PRON – Pronoun
12. PROPN – Proper noun
13. PUNCT – Punctuation
14. SCONJ – Subordinating conjunction
15. SYM – Symbol
16. VERB – Main verb
17. X – Other (foreign words, typos, etc.)

MAIN GOALS OF EVALUATION

- More recent lists of parts-of-speech (or tagsets) have many more word classes –
- 45 for the **Penn Treebank** (Marcus et al., 1993)
- 87 for the **Brown corpus** (Francis, 1979; Francis and Kučera, 1982)
- 146 for the **C7 tagset** (Garside et al., 1997).

USEFULNESS OF POS

- These tagsets distinguish between possessive pronouns (my, your, his, her, its) and personal pronouns (I, you, he, me).
- Knowing whether a word is a possessive pronoun or a personal pronoun can tell us what words are likely to occur in its vicinity (possessive pronouns are likely to be followed by a noun, personal pronouns by a verb).
- This can be useful in a language model for Speech Recognition.

USEFULNESS OF POS

- A word's part-of-speech can tell us something about how the word is pronounced.
- The noun is pronounced CONtent and the adjective conTENT)
- OBject (noun) and obJECT (verb)
- DIScount (noun) and disCOUNT
- Thus knowing the part-of-speech can produce more natural pronunciations in a speech synthesis system and more accuracy in a speech recognition system.

TYPES OF POS TAGGING

- **Rule-based approaches** → Use grammar rules and dictionaries.
- **Statistical models / Probabilistic models** → Hidden Markov Model (HMM), Maximum Entropy (MaxEnt) Tagger, Conditional Random Field (CRF), Maximum Entropy Markov Model (MEMM).
- **Transformation-Based Tagging (TBL)** — also called the **Brill Tagger** — is a hybrid POS tagging approach in NLP that combines the simplicity of rule-based methods with the adaptability of statistical learning.
- **Memory-Based Tagging (MBT)** is an approach to POS tagging in NLP that relies on instance-based (lazy) learning, where the model “remembers” all the training examples and decides the tag for a new word based on its similarity to previously seen contexts.
- **Deep learning approaches** → BiLSTM, Transformers (e.g., BERT).

POS BROAD CATEGORIES

Parts-of-speech can be divided into two broad supercategories:

- **Closed class types:** Closed classes are those that have relatively fixed membership. For example, prepositions are a closed class because there is a fixed set of them in English; new prepositions are rarely coined. Closed class words are also generally function words like of, it, and, or you, which tend to be very short, occur frequently, and often have structuring uses in grammar.
- **Open class types:** By contrast nouns and verbs are open classes because new nouns and verbs are continually coined or borrowed from other languages (e.g., the new verb to fax or the borrowed noun futon). There are four major open classes that occur in the languages of the world; **nouns**, **verbs**, **adjectives**, and **adverbs**. English has all four of these, although not every language does.

NOUN

- The words for most people, places, or things.
- But since syntactic classes like noun are defined syntactically and morphologically rather than semantically, some words for people, places, and things may not be nouns, and conversely some nouns may not be words for people, places, or things.
- Thus nouns include concrete terms like ***ship*** and ***chair***, abstractions like ***bandwidth*** and ***relationship***, and verb-like terms like ***pacing*** as in ***His pacing to and fro became quite annoying.***
- What defines a noun in English, then, are things like its ability to occur with determiners (a goat, its bandwidth, Plato's Republic), to take possessives (IBM's annual revenue), and for most but not all nouns, to occur in the plural form (goats, abaci).

PROPER NOUN

- Nouns are traditionally grouped into **proper nouns** and **common nouns**.
- Proper nouns, like ***Regina***, ***Colorado***, and ***IBM***, are names of specific persons or entities.
- In English, they generally aren't preceded by articles (e.g., ***the book is upstairs, but Regina is upstairs***).
- In written English, proper nouns are usually capitalized.

COMMON NOUN

- In many languages, including English, common nouns are divided into **count nouns** and **mass nouns**.
- Count nouns are those that allow grammatical enumeration; that is, they can occur in both the singular and plural (***goat/goats, relationship/relationships***) and they can be counted (***one goat, two goats***).
- Mass nouns are used when something is conceptualized as a homogeneous group. So words like snow, salt, and communism are not counted (i.e., ****two snows or *two communisms***).
- Mass nouns can also appear without articles where singular count nouns cannot (***Snow is white*** but not ****Goat is white***).

VERB

- The verb class includes most of the words referring to actions and processes, including main verbs like draw, provide, differ, and go.
- English verbs have a number of morphological forms (non-3rd-person-sg (***eat***), 3rd-person-sg (***eats***), progressive (***eating***), past participle (***eaten***)).
- A subclass of English verbs is called **Auxiliaries**.

ADJECTIVE

- The third open class English form is adjectives.
- Semantically this class includes many terms that describe properties or qualities.
- Most languages have adjectives for the concepts of color (***white, black***), age (***old, young***), and value (***good, bad***).
- There are languages without adjectives. In Korean, for example, the words corresponding to English adjectives act as a subclass of verbs, so what is in English an adjective '***beautiful***' acts in Korean like a verb meaning '***to be beautiful***'.

ADVERB

- The final open class form, adverbs, is rather a hodge-podge, both semantically and formally.
- For example Schachter (1985) points out that in a sentence like the following, all the italicized words are adverbs
- Unfortunately, John walked home extremely slowly yesterday
- Directorial adverbs or locative adverbs (*home, here, downhill*) specify the direction or location of some action; degree adverbs (*extremely, very, somewhat*) specify the extent of some action, process, or property; manner adverbs (*slowly, slinkily, delicately*) describe the manner of some action or process; and temporal adverb describe the time that some action or event took place (*yesterday, Monday*).
- Because of the heterogeneous nature of this class, some adverbs (for example temporal adverbs like Monday) are tagged in some tagging schemes as nouns.